

Frequency content of non-periodic signals

Fourier Series (sines, cosines or complex exponentials of discrete frequencies $\omega_0 k$) are used to analyze the frequency content of periodic signals
Fourier Transform is used to analyze the frequency content of non-periodic signal.
Frequency is a continuous parameter.

$$x(t) = \int_{-\infty}^{+\infty} X(f) e^{j2\pi f t} df$$

$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi f t} dt$$

Fourier Transform
Discrete time Fourier Transform
Fast Fourier Transform (for a non-periodic digital signal)
Time is measured in integers (bins)
Frequency is measure in bins too.

$$x[n] = \sum_{k=1}^N X[k] e^{j2\pi \frac{kn}{N}}$$

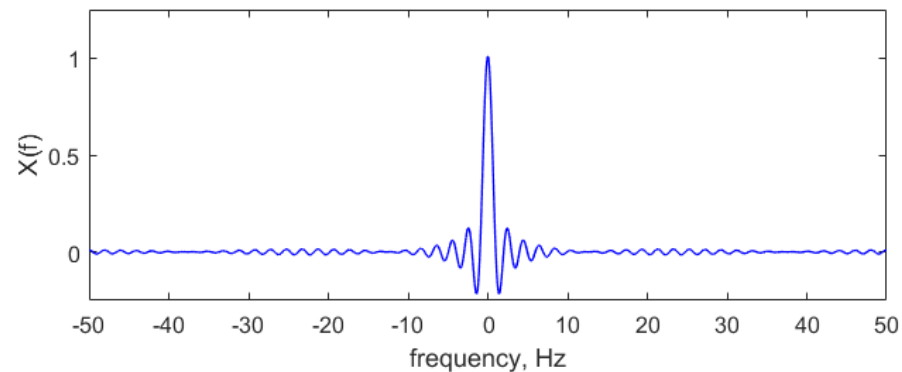
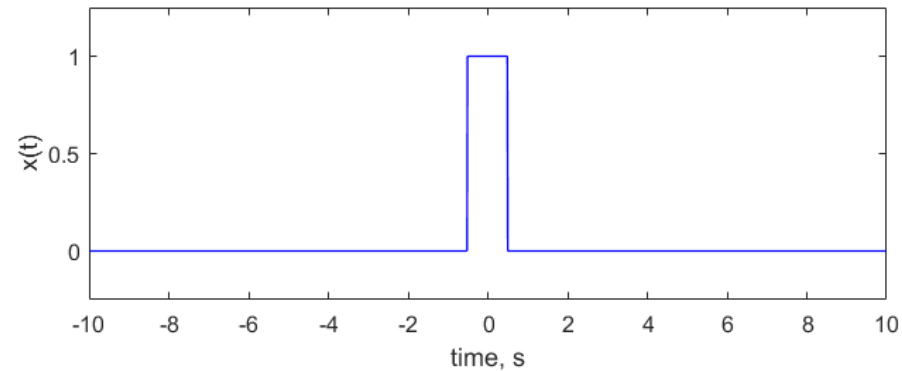
$$X[k] = \frac{1}{N} \sum_{n=1}^N x[n] e^{-j2\pi \frac{kn}{N}}$$

Common Fourier Transform Pairs: Rectangular Pulse

$$\Pi(t) = \begin{cases} 1, & -0.5 < t < 0.5 \\ 0, & \text{otherwise} \end{cases}$$

Fourier transform

$$X(f) = \text{sinc}(f) = \frac{\sin(\pi f)}{\pi f}$$

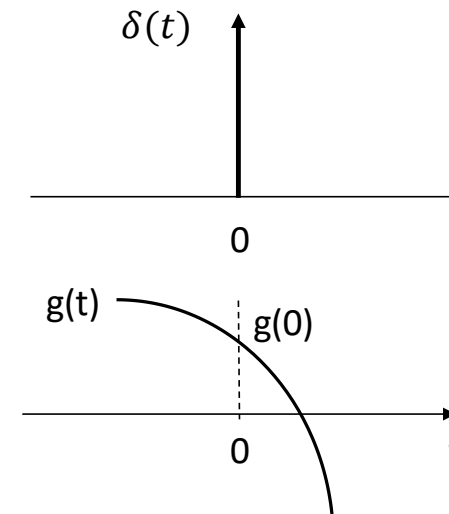


Delta function

Delta function is a mathematical abstraction.
It is used to sample other functions.
Definition:

$$\delta(t) = 0, \text{ for } t \neq 0$$

$$\int_{-\infty}^{+\infty} \delta(t) dt = 1$$



Fourier transform:

$$\int_{-\infty}^{+\infty} \delta(t) e^{j2\pi t f} dt = 1$$

Time delay

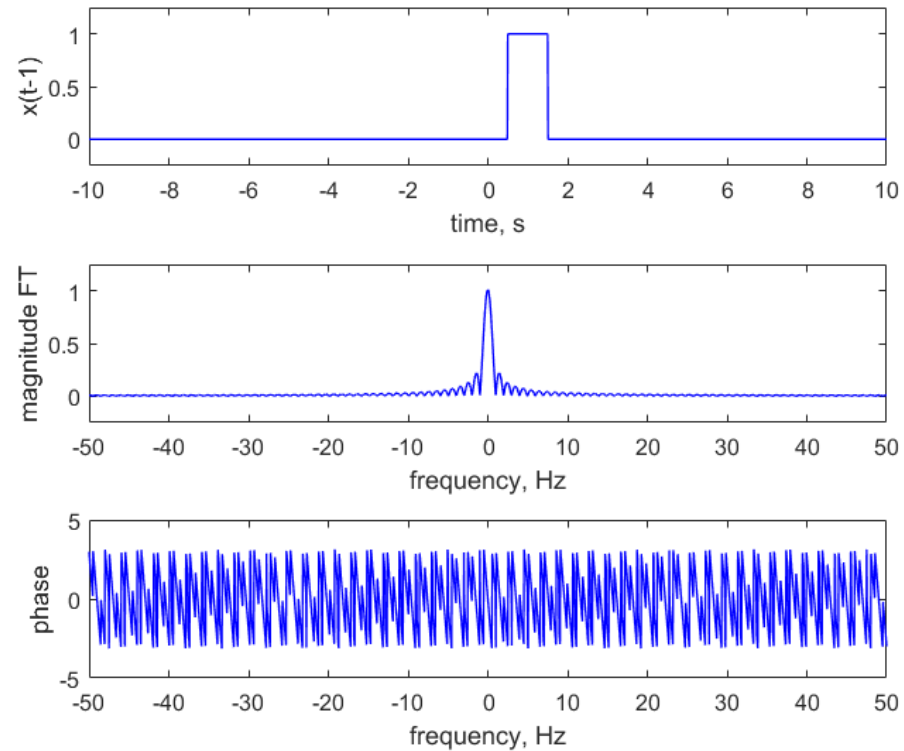
Fourier pairs

$$\Pi(t) \stackrel{FT}{\Leftrightarrow} \text{sinc}(f)$$

$$\Pi(t - 1) \stackrel{FT}{\Leftrightarrow} \text{sinc}(f)e^{-j2\pi f}$$

$$\delta(t) \stackrel{FT}{\Leftrightarrow} 1$$

$$\delta(t - 2) \stackrel{FT}{\Leftrightarrow} e^{-j2\pi f}$$



Linearity of FT

Fourier transform of a sum of signals = sum of Fourier transforms

Example

Complex Exponential and Cosine

Fourier pairs (duality of FT and its inverse)

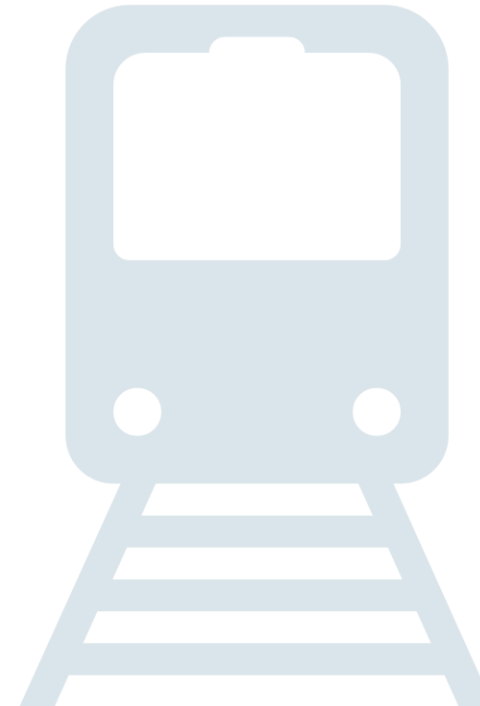
$$e^{j2\pi f_0 t} \stackrel{FT}{\Leftrightarrow} \delta(f - f_0)$$

$$\cos(2\pi f_0 t) = \frac{1}{2} (e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}) \stackrel{FT}{\Leftrightarrow} \frac{1}{2} \delta(f - f_0) + \frac{1}{2} \delta(f + f_0)$$

Fourier transform



Train of pulses: FT of a Dirac train and another Dirac train, but in frequency space.



Correlation

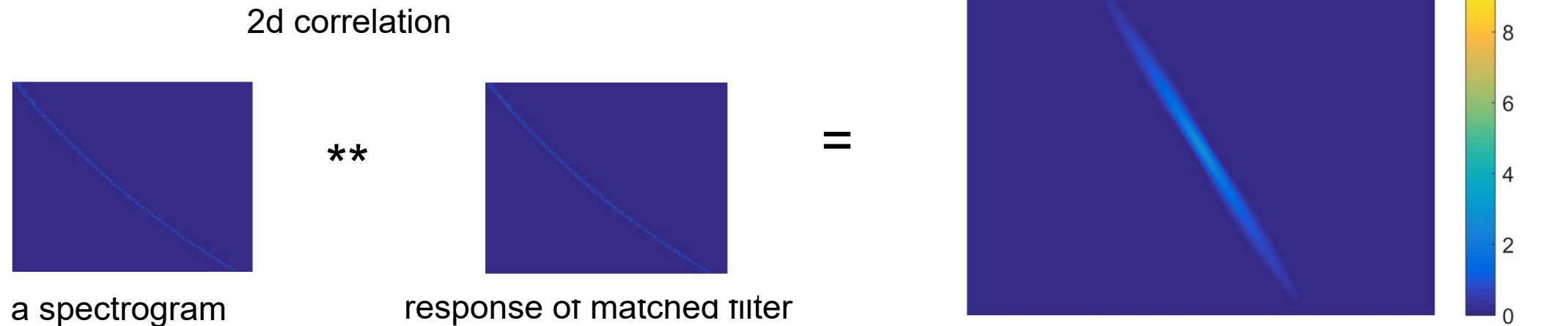
$$g(t) * h(t) \xleftrightarrow{FT} G(f)H(f)^*$$

“*” stands for complex conjugate

No shifting, multiplying and integrating at every t
Only multiplication point-by-point

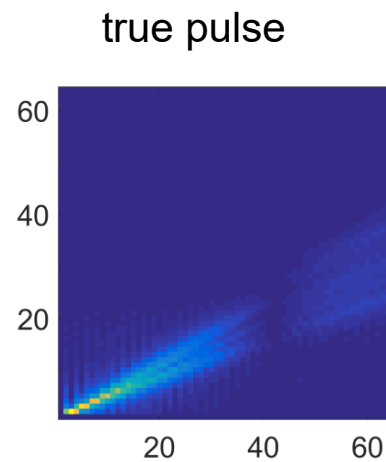
Matched filtering of filter-bank data

Correlate filter-bank data with a dispersed pulse.
Vary pulse width and its shape. Also vary DM value.
Computationally it is very expensive.
PSNR can be used as detection criterion.



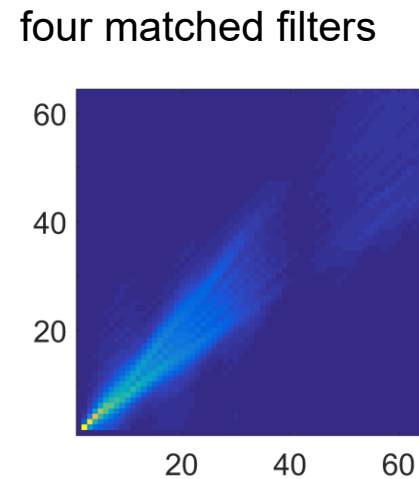
Correlation in 2d FT

2d FT stands for two dimensional Fourier Transform



DM = 50
Width = 100 (in bins)

×



DM = 20, 50, 70, 100
Width = 100 (in bins)

=

the result of “correlation”

